SET - B

Register No															

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
21CSC555J - MACHINE LEARNING ALGORITHM
CYCLE TEST – I

Program Offered: M. Tech-AI&DS | **Year / Semester: I / I**
Max. Marks: 50 | **Duration: 2hr 30 Mins**
Course Code: 21CSC555J | **Date of Exam: 26/10/2024**

Part A (2 x 20 = 30 marks)

Answer any two questions

1 (i) Explain how linear regression works using the mathematical expression for a sample dataset? Also, what are the different performance evaluation metrics, and how do they work? **15**

(ii) Compare Supervised, Unsupervised, and Semi-Supervised Learning. Provide examples of when each type of learning is most suitable. **5**

2 (i). Explain in detail about logistic regression classifier, with suitable example **8**

(ii). Describe the K-fold Cross-Validation process. How does it work, and why is it more robust than a simple train-test split? Provide a scenario where K-fold CV would be useful. **7**

(iii). What is loss function, explain how Gradient descent helps to Minimize the loss **5**

3 (i) How is Logistic Regression extended to handle multi-class classification problems? **10**

(ii) What are the different performance metrics of a Classification model? Explain Confusion matrix, Precision, Recall & F1-Score with example? **10**

Practicals

1 How can we build a logistic regression model using a dummy dataset? Additionally, how do we evaluate the model's performance using classification metrics such as accuracy, precision, recall, and the F1-score? **20**